

Stress-aware P/G TSV Planning in 3D-ICs

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Abstract—Power/Ground (P/G) Through-Silicon-Vias (TSVs) in the Power Distribution Network (PDN) of Three-Dimensional-Integrated-Circuit (3D-IC) have a twofold impact on the delays of the surrounding gates. TSV fabrication causes thermal stress around TSVs, which results in significant carrier mobility variations in their vicinity. On the other hand, the insertion of P/G TSVs will change the voltage of each node in the power grid, which also impacts the delays of the connected gates. Thus, it is necessary to consider the combined effect on delay variation during the P/G TSV planning. In this work, we propose a methodology using Mixed-Integer-Bilinear-Programming (MIBLP) to optimize this delay variation by a refined P/G TSV allocation. Taking into account the impact of thermal stress as well as voltage drop on the circuit delay, we optimally plan the P/G TSVs to minimize the circuit delay for different keep-out zones (KOZs) and PDN pitches.

I. INTRODUCTION

Three-Dimensional-Integrated-Circuit (3D-IC) stacking is a promising technology to continue the integration trend according to the Moore's law. Through-Silicon-Vias (TSVs) are incorporated to establish electrical connections across multiple layers in a 3D chip stack. Benefiting from this technology, the 3D-IC can offer many advantages, such as the reduction in interconnect wire length, improved memory bandwidth, the support for realization of heterogeneous integration, and smaller form factor [1]. However, these advantages come with various costs and issues. Among different challenges, the design of the on-chip power distribution network (PDN) has become a major issue in the 3D-IC design flow. Power/Ground (P/G) TSVs, as an integral part of the 3D PDN, should be optimally planned to realize a reliable power supply design [2], [3], and also for timing optimization.

P/G TSVs in the 3D-IC PDN have a twofold impact on the circuit delay. First, TSV fabrication causes thermal stress around TSVs due to the mismatch in the coefficient of thermal expansion (CTE) between silicon and copper [4]. The thermal stress can cause hole and electron mobility variations in the transistors, which can be translated into variation in the gate delay. For an individual gate, TSV stress-induced timing variations can be as much as $\pm 10\%$ [5]. Moreover, because of larger radiuses of P/G TSVs, compared with signal TSVs, they result in more delay variations [6]. Second, the insertion of P/G TSVs also changes the supply voltage of each node in the PDN [3], impacting the delays of connected gates. The rapid transient current flow in the PDN causes spatial and temporal voltage drops, reducing the supply voltage V_{dd} . The reduced V_{dd} decreases the gate drive strength, thereby increasing the gate delay [7].

Recently, several techniques have explored P/G TSV planning [8]–[11], but most of them focus only on relieving voltage drop in PDN, and the timing optimization is ignored. On the other hand, several methods have studied TSV-induced stress with the focus on the stress-induced timing violations. In [5], a design flow is proposed to analyze timing variations due to TSV-induced stress. In [12], the authors demonstrated that the amount of path delay variations depends on the relative location of the gates to the TSVs, and proposed a set of layout guidelines to optimize delay variations. However, in these work, the TSVs were assumed to be regularly placed in advance.

The timing optimization only relies on the relocations of gates on the critical paths. In [13], a force-directed 3D TSV placement algorithm was proposed, by considering the effect of TSV-induced stress on carrier mobility variation, in which TSV and logic gate locations are determined simultaneously. However, only the impact of TSV-induced stress on circuit delay were considered in these previous work. As an important factor which changes the gate delay, the voltage drop should also be taken into account for timing optimization, especially during the P/G TSV planning.

In the traditional 2D-IC design methodology, PDNs are synthesized early in the design flow. Since the exact placements of gates are not known, designers typically resort to prior experience or rough estimation based on incomplete information. The PDN is created during floorplanning, which is so-called “floorplan and PDN co-synthesis” [14]. In [15], a 3D-IC floorplan and PDN co-synthesis tool was proposed. During the co-synthesis process, the IR-drop and the routing area of the P/G wires are minimized at the same time. However, the P/G TSVs were regularly distributed for each layer due to the incomplete power profile. Compared to the regular one, the irregular TSV distribution topology provides a superiority in terms of power supply noise, TSV number, and footprint area [2], [3], [16]. Therefore, a novel design methodology was proposed in [3], in which the planning of P/G TSVs was performed on a detailed layout. The “post-layout” P/G TSV planning can insert the TSVs into the optimal locations for IR-drop reduction, but also could result in extra delay variation due to the TSV-induced stress and voltage drop, which is ignored in the previous methodologies.

In this paper we propose a stress-aware P/G TSV planning to accurately analyze and then optimize the circuit delay affected by the post-layout insertion of P/G TSVs due to their induced stress and IR-drop. The key contributions of this work include:

- We present a systematic and automatic flow for TSV placement and sizing to optimize P/G TSV planning by considering both voltage- and stress-induced delay variations
- We propose a *Mixed Integer Bilinear Programming* (MIBLP) to optimize the PDN synthesis.

The rest of this paper is organized as follows. First, the compact TSV-induced stress and mobility variation models and 3D PDN design are reviewed in Section II. Section III gives the motivation of our work. Section IV describes our MIBLP-based methodology. In Section V, we report our experimental results. Finally, conclusions are drawn in Section VI.

II. PRELIMINARIES

A. Power delivery in 3D-IC

In 3D-IC, the supply power is fed from the package through power bumps and is distributed over the bottom-most layer via on-chip PDN, which is based on a grid structure [15]. To reach upper layers, the supply power travels through P/G TSVs that connect different layers, as shown in Fig.1. In the PDN, the pitch determines the distance between P/G line, and the width determines the thickness of each

power wire. For each layer in a 3D-IC, there is an individual 2D power grid. P/G TSVs are used to connect the 2D grids of each layer.

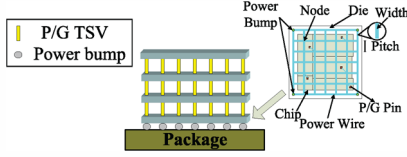


Fig. 1. Three-dimensional power distribution network

B. TSV-induced mobility variation

To take TSV-induced stress into account in the timing optimization, an analytical model of TSV stress-induced carrier mobility variation was proposed in [5]. The mobility change due to thermal stress depends not only on the intensity of the applied stress, but also the orientation between the stress and a transistor channel. If a transistor is located in the overlap zone of multiple TSVs, each TSV works as a stress source. The effect from multiple TSVs can be combined by using linear superposition. Total mobility variation of transistor j ($\Delta\mu_j/\mu_j$) is defined in [5] as

$$\frac{\Delta\mu_j}{\mu_j} = \frac{\Pi B \Delta\alpha \Delta T}{2} \sum_{i \in \text{TSV}(j)} \left(\frac{R_i}{r_{ij}} \right)^2 \alpha(\theta_{ij}) \quad (1)$$

Here Π is a coefficient indicating the sensitivity of mobility change to applied stress, B is biaxial modulus, $\Delta\alpha$ is the CTE difference between copper and silicon, ΔT is the temperature difference between copper annealing and operating temperature. $\text{TSV}(j)$ represents the set of TSVs that influence transistor j , R_i is the radius of TSV i , r_{ij} is a distance from the center of TSV i to the transistor j , and θ_{ij} is defined as the angle between the center of the TSV i and the center of the transistor channel of transistor j when it is placed vertically, as shown in Fig. 2. Note that, for each TSV, there is a KOZ surrounding it. KOZ is the region around the TSV from which all gates must "keep out" so that they are not influenced by the TSV-induced stress to some extent. In realistic designs, the KOZ is defined to ensure that the mobility variation in any transistor around an isolated TSV is below preset tolerance $c\%$ [13].

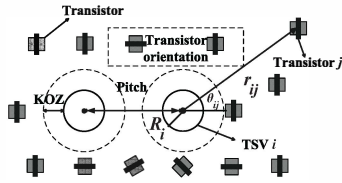


Fig. 2. Illustration of TSV layout parameters.

For the orientation factor α in Equation (1), its value depends on θ and also the type of transistor. The value of θ can be obtained by Equation (2):

$$\theta = \tan^{-1} \left| \frac{X_{TSV} - X_{poly}}{Y_{TSV} - Y_{poly}} \right| \quad (2)$$

where (X_{TSV}, Y_{TSV}) and (X_{poly}, Y_{poly}) indicate the coordinates of TSV center and transistor channel. Fig. 3 gives the values of α for different types of transistors and θ values. As shown in it, we can see that hole mobility increases in a vertical region (mobility enhanced zone), while it decreases in a horizontal direction (mobility degraded zone). 45° has no hole mobility change. For electron mobility variation, horizontal direction has more mobility enhanced zone.

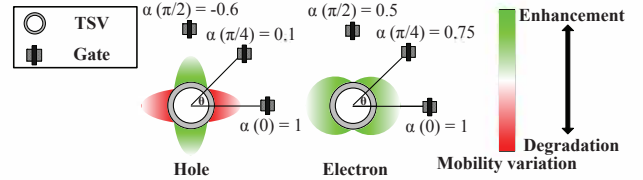


Fig. 3. Carrier mobility variation surrounding TSVs.

C. IR-drop in 3D-IC

In 3D-IC, due to smaller footprints combined with multiple stacked dies, delivering enough current to all components of the 3D stack while guaranteeing power integrity is very challenging. Power integrity consists of low voltage drop in DC components (IR-drop) and low voltage fluctuation in AC components (Ldi/dt). In 3D-IC, IR-drop is particularly important since die stacking has a higher impact on IR-drop than Ldi/dt noise [9]. Thus, we only focus on the IR-drop in this work.

In the IR-drop analysis, the PDN can be modeled as a linear resistive system $\mathbf{G}_{3d}\mathbf{V} = \mathbf{I}$. Here $\mathbf{G}_{3d}$ represents the conductance matrix of the power grid for the interconnected resistors in 3D-IC; $\mathbf{V}$ is a vector containing all the unknown node voltages; and $\mathbf{I}$ is a vector of current loads. Similar to [17], we construct the 3D conductance matrix $\mathbf{G}_{3d}$ using the 2D conductance matrix $\mathbf{G}_{2d}$ and the TSV conductance matrix $\mathbf{G}_{tsv}$. The conductance matrix $\mathbf{G}_{3d}$ of a two-layer 3D-IC, each with a power grid containing N rows and N columns, can be expressed as follows:

$$\mathbf{G}_{3d} = \begin{pmatrix} \mathbf{G}_{2d_1} + \mathbf{G}_{tsv_{1,2}} & -\mathbf{G}_{tsv_{1,2}} \\ -\mathbf{G}_{tsv_{1,2}} & \mathbf{G}_{2d_2} + \mathbf{G}_{tsv_{1,2}} \end{pmatrix} \in \mathbb{R}^{2N^2 \times 2N^2} \quad (3)$$

where $\mathbf{G}_{2d_i} \in \mathbb{R}^{N^2 \times N^2}$ ($i = 1, 2$) is the conductance matrix on layer i without TSVs. $\mathbf{G}_{tsv} \in \mathbb{R}^{N^2 \times N^2}$ represents the conductances due to the existence of TSVs and can be written as:

$$\mathbf{G}_{tsv} = \text{diag}(g_{tsv_1}, g_{tsv_2}, \dots, g_{tsv_m}, \dots, g_{tsv_{N^2}}) \quad (4)$$

where g_{tsv_m} is the conductance of the TSV placed at the m^{th} TSV candidate, which is connected between node m and node $m + N^2$ in the 3D PDN¹. It is similar for multiple layers 3D-IC by apply Equation (3) pair-wisely between adjacent layers [2].

By solving this linear system $\mathbf{G}_{3d}\mathbf{V} = \mathbf{I}$, we can obtain the unknown voltage of each node in the PDN with the current vector $\mathbf{I}$.

III. MOTIVATION

Due to the strong layout dependency, it is possible to take advantage of TSV-induced stress for timing optimization by TSV-stress aware layout optimization, such as gate relocation and TSV placement. In [5], [12], some layout guidelines have been proposed to optimize the stress-induced delay variation with regular TSV distribution topologies. The basic idea is to relocate the gates on the critical and near critical paths in mobility enhanced zone. Fig. 4(a) shows how gate relocation works for timing optimization. Gate 1 is electron critical gate because the timing arc is falling on the path [5]. Therefore, the gate is relocated to be placed close to blue area, which is the electron enhanced zone. However, gate 2 is pulled to hole mobility enhanced zone (i.e., the green area in Fig. 4(a)) because it is hole critical gate. In [13], an irregular stress-aware TSV placement algorithm has been proposed, in which TSV and logic cell locations are determined simultaneously, as shown in Fig. 4(b). It has been

¹In this work, we use the following notation methods to describe a node in a 3D PDN: if a node at the i^{th} row and the j^{th} column in the k^{th} layer power grid, then we notate it as node $(i \cdot N + j + (k - 1) \cdot N^2)$ in the 3D PDN

demonstrated that design with irregular TSV topology has superior performance in terms of wirelength and mobility variation, compared to the regular one [13]. However, the impact of IR-drop on gate delay was ignored in the previous work.

Besides the TSV-induced stress, the IR-drop can also impact the gate delay: a reduced supply voltage deteriorates the gate drive strength, thereby increases the gate delay [7]. As shown in Fig. 4(c), gate 1 is relocated from grid A to grid B following the stress-aware layout optimization, while gate 2 is moved from grid C to grid D. The relocation of gate might change its connected node of the power grid, and hence the supply voltage which drives it. Without considering the negative impact of IR-drop, there is no guarantee that we can benefit from the gate relocation in terms of gate delay. The previous work demonstrate that an optimized P/G TSV planning can relieve IR-drop in 3D PDN [8]–[11], which could also be exploited for timing optimization. In this work, we consider the TSV-induced stress and IR-drop simultaneously during the P/G TSVs planning. With the linear programming (LP) method, we can optimize the circuit delay impacted by P/G TSV insertion.

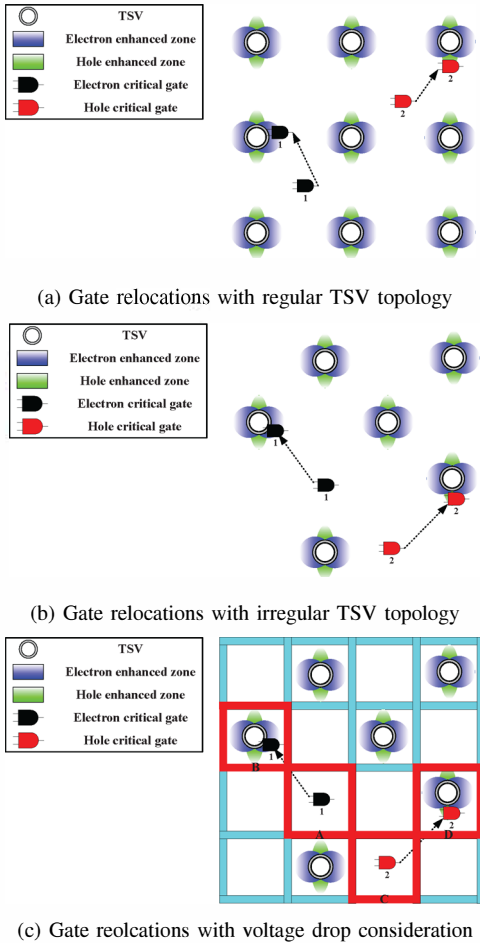


Fig. 4. Gate relocations with regular and irregular TSV topologies

IV. LP OPTIMIZATION FOR P/G TSV PLANNING

A. Problem statement

Our objective is to optimize (minimize) the circuit delay impacted by the insertion of P/G TSVs, subject to: the IR-drop ΔV_{dd} of each node in the PDN be within an acceptable range. Due to the TSV-induced stress and IR-drop, the insertion of P/G TSVs could generate

delay change. The output of this LP optimization is a refined P/G TSV planning (including the location and radius of each inserted TSV), which results in the minimal circuit delay taking the P/G TSV-induced delay change into account.

To accurately take this into consideration, delays of all critical and near-critical paths, i.e. *Vulnerable Paths* (VPs) have to be considered. It should be noted that the P/G TSV-induced delay change of a near-critical path p_i ($\Delta D(p_i)$) could be more than that of a critical path p_j ($\Delta D(p_j)$) such that $D(p_i) + \Delta D(p_i) > D(p_j) + \Delta D(p_j)$. However, in the realistic design, there could be lots of VPs since the paths are fully balanced. Therefore, it is infeasible to take all the VPs into account in the LP formulation. A solution to this issue lies in the selection of only a small set of *Representative Critical Paths* (RCPs) from the numerous VPs [18], [19]. Based on the method proposed in [19], We start the LP formulation with the RCPs, which are selected from VPs. Thus, the LP formulation can be expressed by:

$$\begin{aligned} \text{Minimize : } & \max_{j=0}^N \sum_{\forall g_{ji} \in \text{RCP}_j} (D_0(g_{ji}) + \Delta D(g_{ji})) \\ \text{Subject to : } & \Delta V_{dd}^n \leq \Delta V_{dd}^{\max}, \forall n \end{aligned} \quad (5)$$

where j is an RCP, N is the number of RCPs, and g_{ji} is gate i in RCP $_j$. $D_0(g_{ji})$ is the original delay of gate g_{ji} , and $\Delta D(g_{ji})$ is P/G TSV-induced delay change. ΔV_{dd}^n is the IR-drop of node n in the power grid, and $\Delta V_{dd}^{\max}$ is the acceptable limit of this drop.

B. P/G TSV-induced delay change

In general, the P/G TSV-induced delay change, ΔD , can be divided into two components: a) mobility variation-induced delay change ΔD_{μ} , and b) IR-drop-induced delay change $\Delta D_{V_{dd}}$.

In this work, ΔD can be represented as a sensitivity model:

$$\Delta D = \left(\frac{\partial D}{\partial \mu} \right) \Delta \mu_{TSV} + \left(\frac{\partial D}{\partial V_{dd}} \right) \Delta V_{dd} \quad (6)$$

where $\partial D / \partial \mu$ ($\partial D / \partial V_{dd}$) is the sensitivity of the delay to mobility (supply voltage) changes. Due to the nonlinearity of the sensitivity function, it is stored as a look-up table (LUT) rather than a constant sensitivity value. Here, $\Delta V_{dd} = V_{dd} - V_i$. V_{dd} is the nominal supply voltage, and V_i is the voltage of the node connected by this gate. $\Delta \mu_{TSV}$ is the mobility change due to TSV-induced stress, which is obtained using Equation (1).

In realistic 3D-IC design, TSV radius is an important parameter in this cylindrical via for the designer or manufacturer when considering the TSV density [20]. Moreover, the various radius can also affect the TSV-induced stress [6]. Therefore, we extend Equation (1) to express the P/G TSV-induced mobility variation for each gate j on the RCPs:

$$\begin{aligned} \frac{\Delta \mu_j}{\mu_j} &= \frac{\Pi B \Delta \alpha \Delta T}{2} \sum_{i \in \text{TSV}(j)} \frac{\alpha(\theta_{ij})}{r_{ij}} R_{ik}^2 x_{ik} \\ \sum_{k=1}^n x_{ik} &\leq 1, \quad x_{ik} \in \{0, 1\} \end{aligned} \quad (7)$$

where $\text{TSV}(j)$ is the set of TSV candidates that could influence gate j , working as stress sources. In this work, we assume that there are n different options of TSV radiuses available for the design, and R_{ik} is the radius of the TSV with the k^{th} option. Note that x_{ik} is a binary variable showing if there is a TSV placed in the i^{th} candidate with k^{th} radius option. In this way, we can extract the mobility variation expression for each gate. Based on Equation (5), (6) and (7), we can obtain the objective function by summing all the gate delays on each RPC.

C. Voltage constraints

For each node n in the power grid, the IR-drop ΔV_{dd}^n should be within acceptable limit $\Delta V_{dd}^{\max}$, which is given as

$$\Delta V_{dd}^n = V_{dd} - V_i \leq \Delta V_{dd}^{\max} \quad (8)$$

Here, the voltage variable V_i represents the voltage of each node in the power grid, which can be obtained by solving the linear system $\mathbf{G}_{3d} \mathbf{V} = \mathbf{I}$.

For the matrix $\mathbf{G}_{3d}$, we can construct it as shown in Section II-C. Because of the various radius options of TSV, the conductance of TSV g_{tsv} in Equation (4) is also variable [21]. Similar to Equation (7), the conductance of TSV placed at the i^{th} TSV candidate can be written as:

$$g_{tsv_i} = \sum_{k=1}^n \frac{\pi R_{ik}^2}{\rho h_{tsv}} x_{ik} \quad (9)$$

$$\sum_{k=1}^n x_{ik} \leq 1, \quad x_{ik} \in \{0, 1\}$$

Here ρ is the resistance of the conducting material, and h_{tsv} represents the height of the TSV. Note that we use the same binary variable x to express the various conductance as in Equation (7). Due to the existence of binary variables in $\mathbf{G}_{3d}$, it is very time-consuming to solve this linear system directly. Thus, we use the equations in the system directly as voltage constraints for our LP model.

For the u^{th} equation in this system, we can get the following constraint:

$$\sum_{v=1}^{LN^2} G_{3d_{uv}} V_v = I_u \quad (10)$$

where L is the number of layers in 3D-IC, and N is the number of rows (columns) of power grid for each layer. Here $G_{3d_{uv}}$ is the element at the u^{th} row and the v^{th} column in G_{3d} , V_v is the v^{th} element in $\mathbf{V}$ and I_u is the u^{th} element in $\mathbf{I}$.

Due to the special structure of G_{3d} and the existence of binary variables, there are some nonlinear terms in Equation (10), such as $g_{tsv_v} V_v$, which can be expanded as

$$g_{tsv_v} V_v = g_1 x_{v1} V_v + g_2 x_{v2} V_v + \dots + g_k x_{vk} V_v \quad (11)$$

where g_i ($i = 1, 2, \dots, k$) is conductances of TSVs for different radius. In Equation (11), both x and V are variables. Due to the existence of bilinear terms in the problem, the LP problem is a Mixed-Integer-Bilinear-Linear-Programming (MIBLP) problem. Thus, we need to linearize the bilinear terms first. In this work, we replace them by McCormick envelope [22] to linearize these bilinear terms.

D. Power and stress profiles redistribution

Since we consider the post-layout P/G TSV planning in this work, the proposed approach is performed based on a detailed layout, which means that signal TSVs and the logic gates are already placed. Because signal TSVs have been optimized at their locations to improve wirelength and timing objectives, we avoid overlapping between them. If an optimal P/G TSV location overlaps with gates, we first apply a line-sweeping-based algorithm to find the whitespace [23] in the same layer, and move these gates to the nearest whitespace, which do not overlap with other gates and P/G TSV candidates.

Due to the gate relocations, the power and TSV-induced stress profiles will also be redistributed, which should be taken into account in the LP formulation. For the power consumption, as shown in Fig. 5, due to the insertion of TSV t_i at node i , the overlapped gate a have to be moved to the nearest whitespace, which could change the

power consumptions of related nodes in the power grid. In this case, because of its relocation, gate a will connect to node j , which is the closest node to the gate after the relocation. Thus, there will be a current, which represents the power consumption of gate a , flowing out from node i and flowing into node j .

For the i^{th} TSV candidate, if we insert a P/G TSV with the k^{th} radius option at this location, all the overlapping gates connected to node i and node $i + N^2$ should be relocated, and some of these gates will be connected to another node j in the power grid. We identify these gates, and calculate the "flow-out" and "flow-in" currents in the corresponding nodes, reflected by $-\Delta I_{i,k}$, $-\Delta I_{i+N^2,k}$, and $+\Delta I_{i \rightarrow j,k}$. Then we have

$$\begin{aligned} I_i &= I_i^0 - \Delta I_{i,k} \cdot x_{ik} \\ I_{i+N^2} &= I_{i+N^2}^0 - \Delta I_{i+N^2,k} \cdot x_{ik} \\ I_j &= I_j^0 + \Delta I_{i \rightarrow j,k} \cdot x_{ik} \end{aligned} \quad (12)$$

Here I_i^0 , $I_{i+N^2}^0$ and I_j^0 are the original current sources connected to the nodes i , $i + N^2$ and j , respectively. $-\Delta I_{i,k}$ and $-\Delta I_{i+N^2,k}$ are the currents flowing out from the nodes i and $i + N^2$, while $+\Delta I_{i \rightarrow j,k}$ is the current flowing from node i into node j due to the relocation. In this way, we can handle the issue of power profile redistribution in our LP formulation.

Moreover, if the relocated gate is on an RCP, we will also recalculate the P/G TSV-induced delay change by Equation (6) based on its new location, and update the corresponding term in the objective function. Note that, the nonlinear terms in the objective function, which are generated by the recalculation, can also be linearize by McCormick envelope [22].

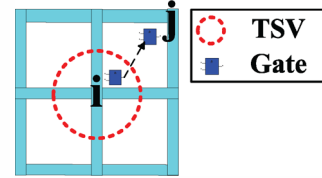


Fig. 5. The redistribution of gates due to the insertion of TSV.

E. Scalability

Scalability has always been a critical issue for LP-based methods [24]. The size of the formulation, which includes the number of variables and constraints, reflects the complexity of LP formulation. As explained below, our approach offers a very good scaling behavior for circuits with increasing gate count.

Suppose we consider a 2D circuit with the number of gates N_{gate} , and partition it into an L -layer 3D-IC. Thus, the number of gates for each layer is roughly N_{gate}/L . Assuming that each node in the power grid can drive s gates at most, the number of P/G TSV candidates, N_{TSV} , is equal to

$$N_{TSV} = \frac{(L-1)}{L} \frac{N_{gate}}{s} \quad (13)$$

If there are k options for the TSV radius, the number of the variables N_{var} in the LP formulation is equal to

$$N_{var} = \left(1 + k \frac{L-1}{L}\right) \frac{N_{gate}}{s} \quad (14)$$

In addition, the number of total constraint N_{constr} is equal to

$$N_{constr} = N_{RCP} + \left(1 + \frac{L-1}{L}\right) \frac{N_{gate}}{s} \quad (15)$$

where N_{RCP} is the number of RCPs. Thus, both the number of variables N_{var} and the number of constraint N_{constr} scale linearly

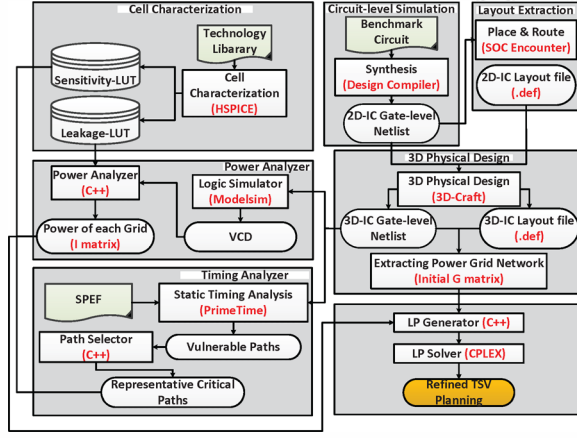


Fig. 6. The detailed implementation flow of the proposed methodology

with the number of gates. Thus, we can expect a good scalability of our formulation for larger circuits.

V. EXPERIMENTAL RESULTS

A. Experimental Setup and Flow

Fig. 6 shows the implementation flow of the proposed MIBLP-based methodology. In our experiment, we assume a three-layer 3D-IC. The benchmark circuits are selected from ITC'99 benchmark suite [25] and the experiments are conducted using the Nangate 45nm library [26]. The experiments were performed on a server with four AMD Opteron 6174 and 256GB RAM. In this work, we assume that a via-last approach is used in 3D-IC fabrication, and we consider three different P/G TSV radiuses: $2.5\mu m$, $5\mu m$, and $10\mu m$ [27]. For the temperature, we assume that ΔT in Equation (7) is $175^\circ C$ in this work, which is the case of $25^\circ C$ for the room temperature and $200^\circ C$ for the copper annealing temperature [28].

As shown in Fig 6, at first, the netlist and the layout of the 2D-ICs are extracted using Synopsys Design Compiler and Cadence SoC Encounter, respectively. UCLA 3D physical design flow "3D-Craft" is exploited to convert the 2D-IC into a 3D-IC [29]. Based on the netlist of each layer, we make a top level netlist that instantiates the layers and connects them using wires which correspond to signal TSV connections. Then we prepare a top level parasitic extraction file (SPEF) for the TSV connections. With a proper timing constraints file, we can run STA (PrimeTime) and get the list of VPs for the 3D-IC. Assisted by the path selector, we select the RCPs from the VPs list.

To obtain the current vector $\mathbf{I}$, we need to calculate the current drawn from each node, by adding the dynamic and leakage currents of all the gates connected to it. For the leakage current, all standard cells in the library are characterized by accurate SPICE simulations to obtain the corresponding leakage current. This information is then stored in leakage Look-Up-Table (LUT). For the dynamic current, we utilize a logic simulator, the circuit netlist, and a set of representative input vector to obtain signal probabilities and activity factors for all the internal nodes in the circuit. Then we send these information into a power analyzer to obtain the power consumptions of each node, and then the current vector $\mathbf{I}$.

In order to generate the objective function in LP formulation, we first get the exact location of each gate on each RCP from the 3D layout, and calculate the mobility variation by Equation (1). By characterizing the standard cell library, we get the mobility and supply voltage sensitivities, and store them in the sensitivity-LUT. Then based on the sensitivity-LUT and RCPs list, we obtain the objective function by Equation (6).

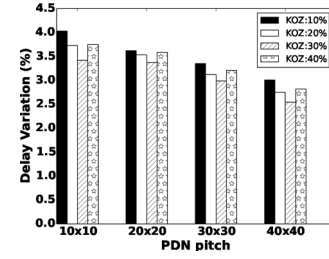


Fig. 7. Impact of KOZ and PDN pitch on induced delay variation

Then, we generate the constraints in the LP formulation. For the linear system $\mathbf{G}_{3d}\mathbf{V} = \mathbf{I}$, the 3D netlist and the 3D layout are used to construct the initial conductance matrix $\mathbf{G}_{3d}$. After calculating the current vector $\mathbf{I}$ by power analyzer, we can obtain the constraints by Equation (10). Finally, the input file for the LP solver is generated and CPLEX [30] is used to obtain the planning of P/G TSVs.

B. Impact of KOZ and PDN pitch on P/G TSV-induced Delay Variation

The KOZ of TSV and the PDN pitch are important 3D-IC PDN parameters, which are worth considering them during the PDN design. To illustrate the impact of KOZ and PDN pitch on the circuit delay, we investigated benchmark b17 with four different KOZs and pitches, respectively. As shown in Fig. 7, the circuit delay variation $\Delta D/D_0 = (D - D_0)/D_0$, increases monotonically with the PDN pitch. For a refined power grid, although ΔD_{μ} might increase due to the increasing number of inserted TSVs, $\Delta D_{V_{dd}}$ could benefit from the refined PDN. However, for the KOZ, delay variation exhibits a "U" shape, and there is an optimal point (in this example at KOZ: 30%). The reason is that although the stress-induced delay change will decrease with the bigger KOZ, the number of TSV candidates will also decrease due to the increased probability overlapping with the fixed signal TSVs, which deteriorates the IR-drop.

C. Comparison with previous technique

To illustrate the significant impact of IR-drop on the circuit delay, we compare our proposed approach with the previous technique, in which only stress-induced delay variation was considered. We investigated a three-layer 30×30 power grid on four different benchmark circuits. Without loss of generality, we set the tolerance value of KOZ $c = 25\%$.

Table I compares the results obtained from our proposed technique to the method presented in [5]. In the previous work [5], the authors claimed that it is possible to take advantage of TSV-induced stress for timing optimization due to its strong layout-dependence. By TSV-stress aware layout optimization, the critical path delay can be reduced. However, because of ignoring the impact of IR-drop on circuit delay, the previous method is too optimistic to overestimate the effects of these methods on the circuit delay optimization. As shown in Table I, the negative values in the fourth column mean that the critical path delay can be reduced by the TSV-stress aware layout optimization without considering the voltage variation. However, the IR-drop, which always has a negative impact on circuit delay, could be more critical compared to the impact of stress variation on the circuit delay. For example, for b17, by taking the voltage variation into account, the circuit delay could be increased to $4.328ns$, while the circuit delay would be reduced to $4.117ns$ without the consideration. The IR-drop can lead to significant (close to 8% in our test case) underestimation of circuit delay.

D. Accuracy and Runtime

Table II shows the results of the proposed MIBLP-based method. As shown in this table, when we consider P/G TSV-induced delay

TABLE II
COMPARISON OF THE ORIGINAL DELAYS WITH OPTIMIZED CIRCUIT DELAYS CONSIDERING P/G TSV-INDUCED DELAY CHANGES

Benchmark Circuit	# of Gates	Original Delay(ns)	PDN	P/G TSV-induced Delay Variation				Runtime (s)
				KOZ : 10% $\Delta D/D_0$	KOZ : 20% $\Delta D/D_0$	KOZ : 30% $\Delta D/D_0$	KOZ : 40% $\Delta D/D_0$	
b17	37117	4.199	10 × 10	4.03%	3.73%	3.42%	3.75%	2.34
			20 × 20	3.62%	3.53%	3.37%	3.59%	15.23
			30 × 30	3.35%	3.12%	2.99%	3.21%	50.27
			40 × 40	3.01%	2.75%	2.54%	2.82%	179.44
b18	92048	11.336	10 × 10	7.95%	7.57%	7.26%	7.51%	2.56
			20 × 20	7.03%	6.34%	6.11%	6.26%	16.77
			30 × 30	5.89%	5.45%	5.02%	5.57%	89.53
			40 × 40	5.31%	5.02%	4.52%	5.15%	143.41
b19	174157	10.057	10 × 10	12.43%	11.49%	10.78%	11.34%	2.77
			20 × 20	11.69%	10.53%	10.04%	10.41%	20.78
			30 × 30	8.57%	7.93%	7.21%	7.47%	100.67
			40 × 40	6.28%	5.75%	5.34%	6.03%	397.46
b22	28317	8.265	10 × 10	8.04%	7.99%	7.32%	7.56%	2.76
			20 × 20	7.21%	6.81%	6.27%	6.73%	15.02
			30 × 30	6.04%	5.71%	5.03%	5.35%	40.78
			40 × 40	5.78%	4.05%	3.51%	3.72%	130.61

TABLE I
COMPARISON WITH PREVIOUS TECHNIQUE (ONLY CONSIDERING TSV-INDUCED STRESS)

Benchmark Circuit	Original Delay(ns)	Only considering stress [5]		Proposed technique	
		D (ns)	$\Delta D/D_0$	D (ns)	$\Delta D/D_0$
b17	4.199	4.117	-1.95%	4.328	3.07%
b18	11.336	11.305	-0.27%	11.946	5.38%
b19	10.057	10.014	-0.43%	10.832	7.71%
b22	8.265	8.281	0.19%	8.719	5.49%

changes, the delay variations for these benchmark circuits show difference varying from 2.54% to 12.43% for different power grids and KOZs. The proposed technique can obtain the minimum circuit delay within a few minutes. Moreover, as shown in Table II, all the delay variations are positive for different PDN pitches and KOZs. These results further prove that the previous work overestimates the benefit for timing optimization produced by gate relocation. Although the TSV-induced stress could improve the circuit delay by gate relocation, IR-drop can cause more degradation on it. Due to the negative impact of IR-drop, the optimized planning of P/G TSVs can only minimize the detrimental effect on circuit delay.

VI. CONCLUSION

Three-dimensional chip stacking with through-silicon-vias (TSVs) has gained tremendous interests for the continuation of the Moore's law. The fabrication of TSVs causes thermal stress, which can lead to significant timing variations. Several technologies have been proposed to take advantage of the TSV-induced stress for timing optimization. However, the negative impact of IR-drop on gate delay was ignored in the previous work. Thus, it is necessary to take the TSV-induced stress and IR-drop into account simultaneously for timing optimization during the P/G TSV planning. In this paper, an MIBLP-based methodology is proposed to solve this optimization problem. According to our simulation results, the combined effects between TSV-induced stress and IR-drop always degrade the circuit delay. By a refined P/G TSV planning, we can minimize the detrimental effect on circuit delay.

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